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Request for Information: Development of an Artificial Intelligence Action Plan

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Comment on FR Doc # 2025-02305

Submitter Information

Email: [REDACTED]

Organization: Association for Intelligent Information Management

General Comment

The Association for Intelligent Information Management (AIIM) is pleased to offer comments in response to the Office of Science and Technology Policy (OSTP)'s request for information regarding an AI Action Plan. Please see our response attached.

Attachments

2025-03 NSF AI Action Plan _ AIIM Response

2025-03 NSF AI Action Plan _ AIIM Response Final



March 10, 2025

Faisal D'Souza
NCO
2415 Eisenhower Avenue
Alexandria, VA 22314, USA

Re: OSTP Request for Information on AI Action Plan

Dear Mr. D'Souza,

The Association for Intelligent Information Management (AIIM) is pleased to offer comments in response to the Office of Science and Technology Policy (OSTP)'s request for information regarding an AI Action Plan.

Founded in 1944, AIIM is a nonprofit organization serving information management practitioners in over 67 countries worldwide. AIIM serves 60,000 community members representing various fields including IT, records management, and knowledge management. Our practitioners focus on the collection, processing, storage, security, retention, and accessibility of unstructured or nontabular data in organizations.

Unstructured data exists outside traditional databases in forms such as documents, records, contracts, invoices, multimedia, social media posts, and emails. While companies historically retained and protected such information mainly for compliance reasons, in the last decade, unstructured data has been increasingly leveraged to improve customer service, enhance decision-making through data analytics, and improve operational efficiency through workflow process automation and artificial intelligence.

As the world's leading association dedicated to the information management, AIIM has extensive experience with AI's implementation in enterprise settings. Unstructured data is vital to training models and completing AI tasks. AIIM helps information leaders manage and prepare unstructured data for AI and automation by providing advice, certification, training, and peer-to-peer support. Through practical and approachable resources, AIIM enables organizations to leverage their information assets effectively, ultimately leading to better business outcomes.

AIIM offers the following response to issues raised by the Office in the notice.

Executive Summary

AIIM recommends that the AI Action Plan prioritize:

1. **Differentiated Regulatory Approach:** Establish a framework that distinguishes between narrow AI applications (which pose limited risks and can be audited) and advanced generative AI systems (which require increased scrutiny).
2. **Federal Framework Development:** Create a flexible, universal AI framework at the federal level that remains adaptable to technological advances, applies across all sectors, and prevents fragmented state-by-state regulation.
3. **Practical Focus Areas:** Emphasize accuracy, transparency, and accountability rather than abstract concepts like "trustworthiness," with clear declaration requirements and source citation standards.
4. **Information Quality Mechanisms:** Support information provenance through source citations, metadata standards, and potentially a "Digital Commons" approach that enhances transparency around AI training and outputs.
5. **Risk Mitigation Strategies:** Address inherent AI risks through mandatory bias testing for high-risk systems, clear remediation requirements, and methods to combat factual inaccuracies and hallucinations.
6. **Human Oversight Integration:** Promote human-in-the-loop approaches as a core principle, with special recognition of information management practitioners as uniquely qualified oversight personnel.
7. **Workforce Development:** Invest in education, collegiate programs, and specialized certification in information management and AI governance to build necessary human oversight capacity.
8. **Advanced AI Architecture Support:** Provide frameworks for responsible innovation in emerging AI architectures (agentic, autonomous, and vertical AI systems) through research funding and regulatory sandboxes.
9. **Responsible Training Data Practices:** Support approaches like standardized metadata automation and content verification mechanisms while recognizing the costs and practical challenges of training data management.

Differentiating AI Risk Levels

A fundamental consideration for any AI policy is the recognition that not all AI systems pose the same level of risk. AIIM strongly recommends classifying AI systems based on their risk profiles:

- **Narrow AI applications** have been used safely for decades, can be audited, and pose limited risks. As noted in our June 2023 comments to NTIA, these systems do not run massive neural networks, and humans have control over what and how the AI learns (i.e., supervised learning). They apply specified or defined rules that have little to no risk associated with them. Examples include document or email readers that can scan the information to either validate the text or sort the item appropriately, technology that can identify and protect sensitive personally identifiable information, or tools that can automate non-sensitive administrative tasks. Both their development processes and training data can be audited and explained.
- **Generative AI and other advanced AI systems** require increased scrutiny and more stringent rules due to higher potential risks. These systems, often referred to as "black-box AI," have internal processes and decision-making mechanisms that cannot be determined or audited, making it difficult to hold it and the entity using it accountable for the decisions made.

This differentiated approach will prevent overregulation of low-risk AI while ensuring appropriate guardrails for higher-risk systems. Narrow AI should not be regulated in the same manner as Generative AI.

Federal Framework Priorities

AIIM urges OSTP to prioritize developing a flexible, universal AI framework at the federal level. The current regulatory uncertainty has significantly hindered innovation and AI adoption across industries. As we noted in our June 2023 response to NTIA, "organizations are reluctant to implement available AI technologies due to legitimate concerns about potential unintended harm, liability risks, and uncertain regulatory consequences."

The lack of public policy strategy or legal framework has resulted in U.S. end user companies not keeping pace with the expansion and availability of AI. To boost and maintain national competitiveness, AIIM recommends the U.S. government provide clear policies and legal frameworks to reduce perceived and real liabilities and empower end user companies to invest in AI.

This framework should incorporate successful principles from established AI policies in other jurisdictions while addressing unique U.S. considerations. AIIM strongly recommends that U.S. states be encouraged to align their policies with this federal framework rather than creating disparate regulations.

The business case for regulatory harmonization is compelling. From 2018-2021, organizations worldwide faced substantial challenges complying with the fragmented landscape of data privacy regulations, including the EU's General Data Protection Regulation (GDPR) implemented in 2018 and the California Consumer Privacy Act (CCPA) effective in 2020. While these regulations served important consumer protection goals, their inconsistent requirements created significant compliance burdens, particularly for small and medium-sized enterprises operating across multiple jurisdictions.

A coordinated federal AI framework would prevent repeating this fragmented regulatory experience, fostering responsible innovation while providing clear, consistent guidance for organizations developing and deploying AI technologies.

This framework should:

1. **Be universally applicable across sectors.** The framework must be universally applied to every sector and industry. Attempting to design a framework for each sector or industry would be overly complicated and unnecessarily burdensome. It may also result in policy that does not easily adapt to changing technology or circumstances or growth of the technology into new areas of the economy. It's important for ease of business and consistency that regulations are universal. Businesses of all sizes have experienced the challenge of data privacy laws, which vary by country and even state/territory.
2. **Remain flexible to accommodate the rapid pace of AI development.** AI technology has advanced exponentially, and deployment has accelerated to the point that AI is pervasive throughout our economy, our culture, and our day-to-day lives. The government simply cannot keep pace, and any rigid public policy that includes strict requirements would be obsolete in a matter of months, if not weeks or even days.
3. **Be implemented at the federal level to prevent an unworkable regulatory landscape.** If states regulate AI in their own manner, entities will be forced to abide by and monitor inevitable policy changes on a state-by-state basis. This patchwork would be impossible to implement, and compliance would be unmanageable and too costly for private businesses.
4. **Focus on principles rather than specific technical requirements** that would quickly become obsolete. It is critical that the government not attempt to implement regulations but instead focus on a framework that outlines key principles and characteristics that will aid entities in identifying appropriate AI.
5. **Leverage existing frameworks** like NIST frameworks without making them overly rigid or mandatory. The National Institute for Standards and Technology's ('NIST') Cybersecurity Framework is a good model to follow, but AIIM cautions that this framework should not be adopted as regulation. Instead, existing standards should be used as a guidepost for the regulated community that can grow and adapt to new technology or advancements.

Accuracy, Transparency, and Accountability

The AI Action Plan should focus on practical goals rather than abstract concepts. As we stated in our June 2023 NTIA comments, the focus of policy should be "on accuracy of AI technology, not obtaining the potentially unattainable 'trustworthiness.' Attempting to achieve this abstract goal would be a nearly impossible task and would only result in the government getting bogged down in a quagmire."

Instead, policy should focus on:

1. **Accuracy:** As AIIM noted in our June 2023 NTIA comments, accuracy is paramount in AI systems. Users need assurance that information generated by AI is credible, current, and complete. When users can verify the reliability of AI outputs, they can make informed decisions and hold entities accountable for inaccurate or misleading information. Policy should prioritize mechanisms that enable verification of information quality.
2. **Transparency:** AIIM has consistently advocated that transparency is essential for public trust in AI systems. Consumers deserve to know when AI is being used to make decisions that affect them, how those systems operate, and why specific approaches were chosen. Without this transparency, meaningful consent and informed use become impossible.
3. **Accountability:** AIIM maintains that accountability must flow to both AI developers and the entities implementing AI systems. Organizations deploying AI technologies must take responsibility for their implementation choices, while their employees must adhere to proper usage guidelines. When entities purchase AI tools, they assume responsibility for managing the associated risks and ensuring appropriate deployment.

AIIM recommends exploring:

1. Declaration requirements to inform consumers when AI is used to make decisions and how those decisions were made. Knowing AI is used and how it is being used will be key in promoting AI usage and ensuring accountability.
2. Source citation and metadata standards for AI outputs, which AIIM identified in our October 2023 comments to the U.S. Copyright Office as crucial for establishing information provenance and to mitigate the risks of distributing and employing inaccurate, unverified, or unlawful information.

Information Provenance and Quality

For AI systems to be trusted and beneficial, the Action Plan should address information provenance. This requires robust approaches to information quality:

1. **Source citations** are an important way to establish information provenance and to mitigate the risks of distributing and employing inaccurate, unverified, or unlawful information. AIIM believes that if the origins of AI-generated content are maintained by

developers, the training data would be more transparent. It would then make it easier to be verified later in the iterative path—and protect end-user businesses.

2. **Metadata standards** can facilitate identification of AI-generated content. Metadata and metadata automation could be used to facilitate an efficient opt-out approach that occurs before data is mined for training models. AIIM recommends an approach that currently web search engines follow where the robots.txt metadata file indicates whether it's acceptable for a website's content to be indexed or not.
3. A **"Digital Commons" approach** could enhance transparency. As we've advocated in our Copyright Office comments, publicly accessible disclosures could be a part of a 'Digital Commons', which would create greater transparency and trust around how copyrighted information was used.

AIIM maintains that three proven, technically feasible approaches would help mitigate risks while supporting innovation:

1. Source citations with metadata in AI outputs that allow users to trace and verify information origins
2. Digital watermarking of AI-generated content to clearly indicate its synthetic nature
3. Confidence rating systems that provide users with an assessment of the potential accuracy of AI outputs

These approaches, which we outlined in our PCAST Working Group response, balance the need for transparency with practical implementation considerations.

Addressing Inherent Risks in AI Systems

The AI Action Plan must go beyond traditional regulatory concerns like privacy, security, and intellectual property to address risks inherent to the technology itself. As information management practitioners, AIIM members are particularly concerned with issues of accuracy, bias, and the potential for harm from AI systems.

Bias Testing and Remediation

AI systems trained on historical data often replicate and amplify existing societal biases. The Action Plan should prioritize:

- Mandatory bias testing for high-risk AI systems that make or inform decisions affecting individual rights, opportunities, or access to essential services
- Clear methodologies for identifying and measuring various forms of bias, including those related to race, gender, age, and other protected characteristics
- Remediation requirements when bias is detected, with objective benchmarks for improvement
- Regular auditing and reporting on bias metrics for systems in ongoing use

Bias detection and remediation should be conducted throughout the AI lifecycle, from dataset compilation through model training and into production deployment. These requirements should be proportional to the potential risk of harm, with higher standards for systems used in critical domains such as healthcare, criminal justice, lending, and employment.

Factual Accuracy and Hallucination Mitigation

Generative AI systems are prone to producing plausible-sounding but factually incorrect information—a phenomenon often called "hallucination." The Action Plan should address this by:

- Encouraging development of reliable methods to assess factual accuracy in AI outputs
- Supporting research into technical approaches for reducing hallucinations
- Promoting transparency about known limitations in factual reliability
- Requiring clear disclaimers when AI systems generate content that appears factual but may contain errors

Information management best practices, including validation against authoritative sources and proper documentation of information provenance, are essential for addressing these accuracy issues.

Human-in-the-Loop as a Core Principle

AIIM strongly recommends that the AI Action Plan establish human oversight as a fundamental principle for responsible AI deployment. Human-in-the-loop (HITL) approaches provide critical safeguards against the limitations and risks of fully automated systems.

Information Management Practitioners as Ideal Oversight Personnel

Information management practitioners are uniquely positioned to serve as effective human reviewers in AI systems. With decades of experience managing unstructured data—which constitutes the majority of information that generative AI systems process and create—these practitioners already possess critical skills for AI oversight:

- Deep expertise in information validation, quality control, and verification
- Understanding of metadata standards and information classification
- Experience with information lifecycle management and governance
- Ability to assess information provenance and reliability
- Familiarity with compliance requirements across multiple domains

The Action Plan should recognize that information management practitioners represent an existing workforce with transferable skills ideally suited for AI oversight roles. Rather than creating entirely new professional categories, the administration should leverage and enhance this established expertise.

Essential Elements of Human Oversight

The Action Plan should promote HITL design that includes:

1. Meaningful human review of high-consequence AI decisions before implementation
2. Clear procedures for humans to override AI recommendations when appropriate
3. Balanced workloads that allow human reviewers sufficient time for thoughtful assessment
4. Training programs that prepare human operators to effectively identify and address AI errors
5. Authority structures that empower human reviewers to escalate concerns

Human oversight should be calibrated to the potential risk and impact of AI systems. While low-risk applications may require minimal human involvement, high-risk contexts demand substantial human oversight. The default approach should always include human review for novel or consequential applications, with exceptions being rare and thoroughly justified.

Educational Investment and Workforce Development

To build the necessary human oversight capacity, the Action Plan should:

1. Incentivize collegiate programs in information management and information governance
2. Support post-graduate workforce training in AI oversight methods
3. Support industry-provided, specialized certification programs for professional providing AI oversight
4. Create tax incentives for organizations that invest in training employees for AI oversight roles
5. Fund research into effective human-AI collaboration methodologies

AIIM recommends that the administration establish grant programs specifically targeting educational institutions that incorporate information management and AI governance into their curricula. Additionally, professional development initiatives should be created to help existing information professionals adapt their skills to AI oversight contexts.

The United States currently has a shortage of qualified personnel for effective AI oversight. By investing in education and training focused on unstructured data management—the foundation of modern AI systems—the administration can address this gap while creating high-value employment opportunities.

Supporting Advanced AI Architectures

The AI Action Plan should anticipate the evolution of AI technologies and provide a supportive framework for responsible innovation in advanced AI architectures. Future economic growth and U.S. competitiveness will increasingly depend on cutting-edge AI implementations.

Agentic, Autonomous, and Vertical AI Systems

AIIM recommends that the Action Plan specifically address:

- **Agentic AI:** Systems that can act independently to achieve specified goals require specialized governance approaches. The Action Plan should support research into safe deployment models, effective oversight mechanisms, and clear lines of accountability for agentic systems.
- **Autonomous AI:** As AI systems become increasingly capable of operating without direct human supervision, the Plan should address appropriate safety measures, performance monitoring, and intervention protocols. These systems offer significant efficiency benefits but require carefully designed guardrails.
- **Vertical AI:** Domain-specific AI systems optimized for particular industries or functions represent a major growth area for enterprise adoption. The Plan should encourage the development of industry-specific best practices and standards for responsible vertical AI applications.

These advanced architectures are likely to drive the next wave of enterprise AI adoption and economic productivity gains. By providing clear guidance for their responsible development and deployment, the Action Plan can position U.S. businesses for leadership in these emerging technologies.

Research and Development Support

To maintain U.S. leadership in advanced AI architectures, the Action Plan should:

1. Direct federal research funding toward responsible deployment mechanisms for advanced AI systems
2. Establish collaborative research initiatives between government, industry, and academia
3. Create regulatory sandboxes that allow controlled testing of innovative approaches
4. Develop specialized governance frameworks for different categories of advanced AI

By taking a forward-looking approach to these emerging technologies, the Action Plan can ensure that regulatory frameworks evolve alongside technological capabilities, providing appropriate guardrails without stifling innovation.

Supporting Responsible AI Training

To support high-quality AI development, the Action Plan should address concerns around AI training data and intellectual property. AIIM recommends practical approaches consistent with our previous positions:

1. **Leveraging approaches similar to robots.txt for content creators.** AIIM advocates for standardization and leveraging existing technology and processes to ease the burden of implementation. In our Copyright Office comments, "AIIM recommends an approach that currently web search engines follow where the robots.txt metadata file indicates

whether it's acceptable for a website's content to be indexed or not. Many AI models based on publicly available datasets could draw upon such an approach."

2. **Supporting standardization of metadata automation for data provenance.** Interoperability of various record management or content management systems is a significant obstacle. Metadata from one system would need to be mapped to the metadata of the receiving system for opt outs to be recognized. Industry standards can help establish this, but standards are not mandatory without regulation and legislation.
3. **Recognizing the substantial costs of record retention for training data.** There are costs for developing and maintaining a record management program. For an AI developer to retain training materials at-scale, they would need to employ the principles of records management and would likely need to adopt enterprise systems, like Records Management systems to not only retain records, but also be able to comply with regulatory requirements and possible legal requests. While AIIM avidly supports records retention in this instance, it's important to understand the impact on and costs to business when considering regulations.
4. **Encouraging approaches that enable content verification rather than requiring licensing.** AIIM's recommendation would be to mandate the inclusion of source information as metadata as part of legislation as opposed to requiring licensure to use content. Restrictions on content stymie innovation and human creativity. Generative AI has the potential to build upon existing human knowledge at a rapid scale by connecting and combining data from a vast number of sources. Licensure could slow the potential of generative AI whereas source citations would support and grow use of generative AI by building trust in AI outputs amongst users while providing credit to the original copyright owners.

Conclusion

AIIM believes that with appropriate policies, the United States can maintain leadership in AI development while ensuring that these technologies are developed and deployed responsibly.

The key is to establish principles that encourage innovation while providing the certainty businesses need to confidently adopt and implement AI technologies. Government's role should be to incentivize adoption of accountable, responsible AI. A flexible, universal framework applied at the federal level would provide entities with the information and surety they need to adopt the technology and fully incorporate it into their operations soundly and safely.

The Association for Intelligent Information Management, with our deep expertise in information management principles and practices across multiple industries, stands ready to assist OSTP in developing effective AI policies that maintain U.S. leadership while promoting responsible use.

We look forward to working with you and in collaboration with our member organizations and other stakeholders to support AI innovation and growth while balancing innovation with risk mitigation.

Sincerely,

Tori Miller Liu, MBA, FASAE, CAE, CIP
President & CEO
Association for Intelligent Information Management (AIIM)
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Works Cited

- Liu, Tori. Association for Intelligent Information Management. Comment ID NTIA-2023-0005-1188. Comment on AI Accountability Policy Request for Comment. National Telecommunications and Information Administration, Docket No. [REDACTED] June 12, 2023. <https://www.regulations.gov/comment/NTIA-2023-0005-1188>
- Liu, Tori. Association for Intelligent Information Management (AIIM). Comment ID COLC-2023-0006-8502. Comment on Artificial Intelligence and Copyright. Library of Congress, Copyright Office, Docket No. 2023-6; COLC-2023-0006. October 29, 2023. <https://www.regulations.gov/comment/COLC-2023-0006-8502>
- Liu, Tori. Association for Intelligent Information Management (AIIM). Comment on PCAST Working Group on Generative AI Invites Public Input. President's Council of Advisors on Science and Technology. July 28, 2023.



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2. **Remain flexible to accommodate the rapid pace of AI development.** AI technology has advanced exponentially, and deployment has accelerated to the point that AI is pervasive throughout our economy, our culture, and our day-to-day lives. The government simply cannot keep pace, and any rigid public policy that includes strict requirements would be obsolete in a matter of months, if not weeks or even days.
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developers, the training data would be more transparent. It would then make it easier to be verified later in the iterative path—and protect end-user businesses.

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- Promoting transparency about known limitations in factual reliability
- Requiring clear disclaimers when AI systems generate content that appears factual but may contain errors

Information management best practices, including validation against authoritative sources and proper documentation of information provenance, are essential for addressing these accuracy issues.

Human-in-the-Loop as a Core Principle

AIIM strongly recommends that the AI Action Plan establish human oversight as a fundamental principle for responsible AI deployment. Human-in-the-loop (HITL) approaches provide critical safeguards against the limitations and risks of fully automated systems.

Information Management Practitioners as Ideal Oversight Personnel

Information management practitioners are uniquely positioned to serve as effective human reviewers in AI systems. With decades of experience managing unstructured data—which constitutes the majority of information that generative AI systems process and create—these practitioners already possess critical skills for AI oversight:

- Deep expertise in information validation, quality control, and verification
- Understanding of metadata standards and information classification
- Experience with information lifecycle management and governance
- Ability to assess information provenance and reliability
- Familiarity with compliance requirements across multiple domains

The Action Plan should recognize that information management practitioners represent an existing workforce with transferable skills ideally suited for AI oversight roles. Rather than creating entirely new professional categories, the administration should leverage and enhance this established expertise.

Essential Elements of Human Oversight

The Action Plan should promote HITL design that includes:

1. Meaningful human review of high-consequence AI decisions before implementation
2. Clear procedures for humans to override AI recommendations when appropriate
3. Balanced workloads that allow human reviewers sufficient time for thoughtful assessment
4. Training programs that prepare human operators to effectively identify and address AI errors
5. Authority structures that empower human reviewers to escalate concerns

Human oversight should be calibrated to the potential risk and impact of AI systems. While low-risk applications may require minimal human involvement, high-risk contexts demand substantial human oversight. The default approach should always include human review for novel or consequential applications, with exceptions being rare and thoroughly justified.

Educational Investment and Workforce Development

To build the necessary human oversight capacity, the Action Plan should:

1. Incentivize collegiate programs in information management and information governance
2. Support post-graduate workforce training in AI oversight methods
3. Support industry-provided, specialized certification programs for professional providing AI oversight
4. Create tax incentives for organizations that invest in training employees for AI oversight roles
5. Fund research into effective human-AI collaboration methodologies

AIIM recommends that the administration establish grant programs specifically targeting educational institutions that incorporate information management and AI governance into their curricula. Additionally, professional development initiatives should be created to help existing information professionals adapt their skills to AI oversight contexts.

The United States currently has a shortage of qualified personnel for effective AI oversight. By investing in education and training focused on unstructured data management—the foundation of modern AI systems—the administration can address this gap while creating high-value employment opportunities.

Supporting Advanced AI Architectures

The AI Action Plan should anticipate the evolution of AI technologies and provide a supportive framework for responsible innovation in advanced AI architectures. Future economic growth and U.S. competitiveness will increasingly depend on cutting-edge AI implementations.

Agentic, Autonomous, and Vertical AI Systems

AIIM recommends that the Action Plan specifically address:

- **Agentic AI:** Systems that can act independently to achieve specified goals require specialized governance approaches. The Action Plan should support research into safe deployment models, effective oversight mechanisms, and clear lines of accountability for agentic systems.
- **Autonomous AI:** As AI systems become increasingly capable of operating without direct human supervision, the Plan should address appropriate safety measures, performance monitoring, and intervention protocols. These systems offer significant efficiency benefits but require carefully designed guardrails.
- **Vertical AI:** Domain-specific AI systems optimized for particular industries or functions represent a major growth area for enterprise adoption. The Plan should encourage the development of industry-specific best practices and standards for responsible vertical AI applications.

These advanced architectures are likely to drive the next wave of enterprise AI adoption and economic productivity gains. By providing clear guidance for their responsible development and deployment, the Action Plan can position U.S. businesses for leadership in these emerging technologies.

Research and Development Support

To maintain U.S. leadership in advanced AI architectures, the Action Plan should:

1. Direct federal research funding toward responsible deployment mechanisms for advanced AI systems
2. Establish collaborative research initiatives between government, industry, and academia
3. Create regulatory sandboxes that allow controlled testing of innovative approaches
4. Develop specialized governance frameworks for different categories of advanced AI

By taking a forward-looking approach to these emerging technologies, the Action Plan can ensure that regulatory frameworks evolve alongside technological capabilities, providing appropriate guardrails without stifling innovation.

Supporting Responsible AI Training

To support high-quality AI development, the Action Plan should address concerns around AI training data and intellectual property. AIIM recommends practical approaches consistent with our previous positions:

1. **Leveraging approaches similar to robots.txt for content creators.** AIIM advocates for standardization and leveraging existing technology and processes to ease the burden of implementation. In our Copyright Office comments, "AIIM recommends an approach that currently web search engines follow where the robots.txt metadata file indicates

whether it's acceptable for a website's content to be indexed or not. Many AI models based on publicly available datasets could draw upon such an approach."

2. **Supporting standardization of metadata automation for data provenance.** Interoperability of various record management or content management systems is a significant obstacle. Metadata from one system would need to be mapped to the metadata of the receiving system for opt outs to be recognized. Industry standards can help establish this, but standards are not mandatory without regulation and legislation.
3. **Recognizing the substantial costs of record retention for training data.** There are costs for developing and maintaining a record management program. For an AI developer to retain training materials at-scale, they would need to employ the principles of records management and would likely need to adopt enterprise systems, like Records Management systems to not only retain records, but also be able to comply with regulatory requirements and possible legal requests. While AIIM avidly supports records retention in this instance, it's important to understand the impact on and costs to business when considering regulations.
4. **Encouraging approaches that enable content verification rather than requiring licensing.** AIIM's recommendation would be to mandate the inclusion of source information as metadata as part of legislation as opposed to requiring licensure to use content. Restrictions on content stymie innovation and human creativity. Generative AI has the potential to build upon existing human knowledge at a rapid scale by connecting and combining data from a vast number of sources. Licensure could slow the potential of generative AI whereas source citations would support and grow use of generative AI by building trust in AI outputs amongst users while providing credit to the original copyright owners.

Conclusion

AIIM believes that with appropriate policies, the United States can maintain leadership in AI development while ensuring that these technologies are developed and deployed responsibly.

The key is to establish principles that encourage innovation while providing the certainty businesses need to confidently adopt and implement AI technologies. Government's role should be to incentivize adoption of accountable, responsible AI. A flexible, universal framework applied at the federal level would provide entities with the information and surety they need to adopt the technology and fully incorporate it into their operations soundly and safely.

The Association for Intelligent Information Management, with our deep expertise in information management principles and practices across multiple industries, stands ready to assist OSTP in developing effective AI policies that maintain U.S. leadership while promoting responsible use.

We look forward to working with you and in collaboration with our member organizations and other stakeholders to support AI innovation and growth while balancing innovation with risk mitigation.

Sincerely,

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Works Cited

- Liu, Tori. Association for Intelligent Information Management. Comment ID NTIA-2023-0005-1188. Comment on AI Accountability Policy Request for Comment. National Telecommunications and Information Administration, Docket No. [REDACTED] June 12, 2023. <https://www.regulations.gov/comment/NTIA-2023-0005-1188>
- Liu, Tori. Association for Intelligent Information Management (AIIM). Comment ID COLC-2023-0006-8502. Comment on Artificial Intelligence and Copyright. Library of Congress, Copyright Office, Docket No. 2023-6; COLC-2023-0006. October 29, 2023. <https://www.regulations.gov/comment/COLC-2023-0006-8502>
- Liu, Tori. Association for Intelligent Information Management (AIIM). Comment on PCAST Working Group on Generative AI Invites Public Input. President's Council of Advisors on Science and Technology. July 28, 2023.